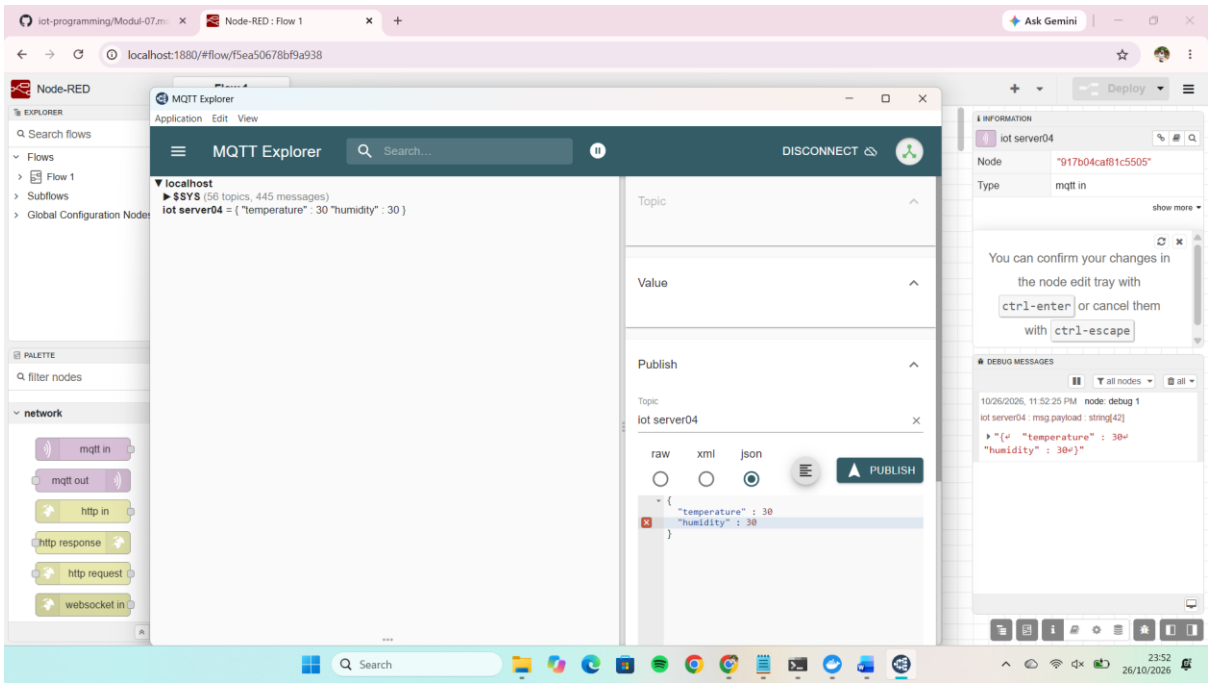
LAPORAN PRAKTIKUM MODUL 7

Nama: Novita Syifa Azahra

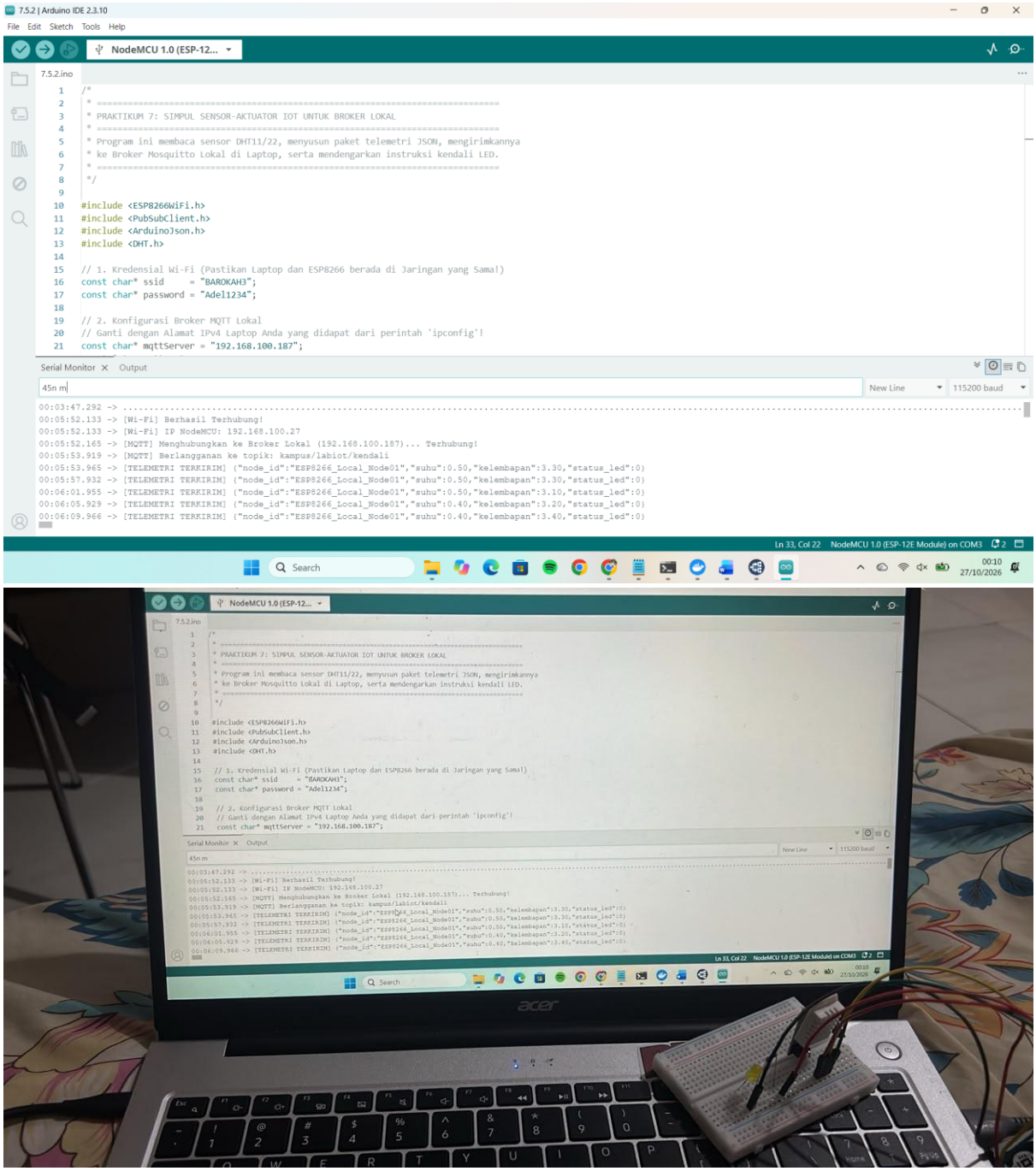
NPM: 25782078

7.5.1 Praktikum 1: Pembangunan Ekosistem IoT Server dengan Docker Compose

```
PS D:\DOCKER01\iot-server04> docker compose up -d
[+] up 2/2
 ✔Container mqtt-broker Started                                                    0.5s
 ✔Container node-red    Started                                                    0.6s
PS D:\DOCKER01\iot-server04> docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                            PORTS
                                                                                                                          NAMES
88dfa0fde31f   nodered/node-red:latest        "./entrypoint.sh"        26 seconds ago   Up 25 seconds (health: starting)  0.0.0.0:1880->1880/tcp, [::]:1880->18
80/tcp                                                                                                                    node-red
e335c8626dcb   eclipse-mosquitto:latest       "/docker-entrypoint.…"   26 seconds ago   Up 26 seconds                     0.0.0.0:1883->1883/tcp, [::]:1883->18
83/tcp, 0.0.0.0:9001->9001/tcp, [::]:9001->9001/tcp   mqtt-broker
PS D:\DOCKER01\iot-server04>
```

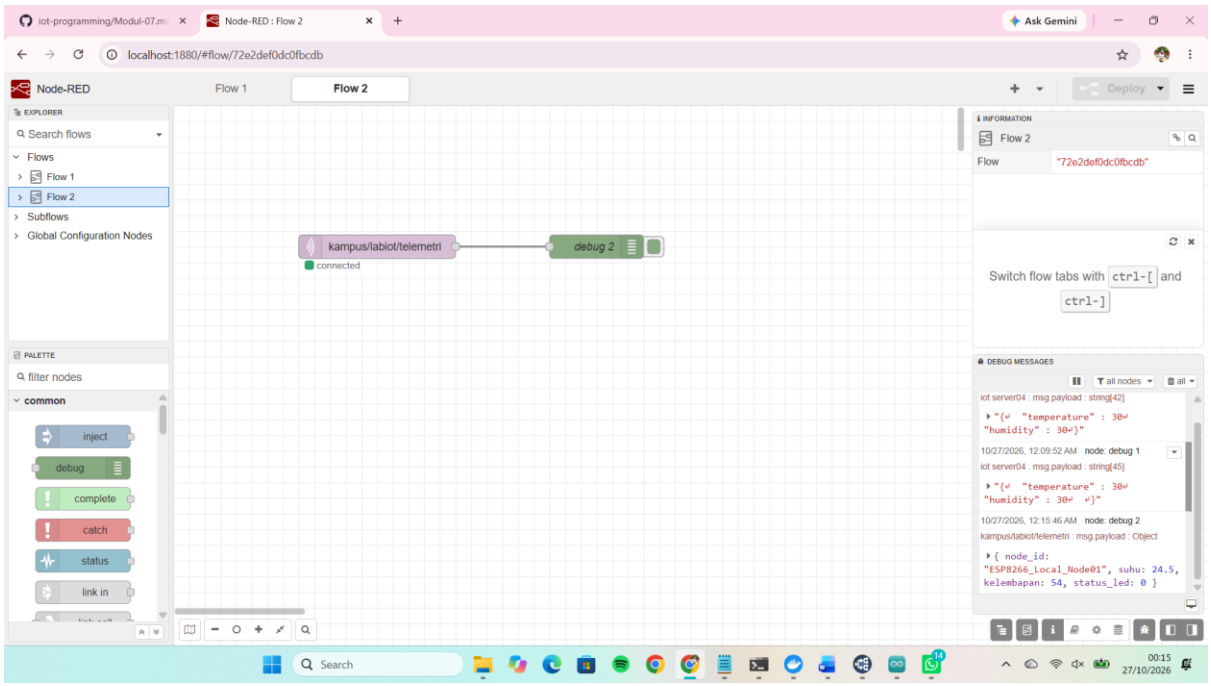


7.5.2 Praktikum 2: Migrasi Simpul NodeMCU ESP8266 ke Broker MQTT Lokal

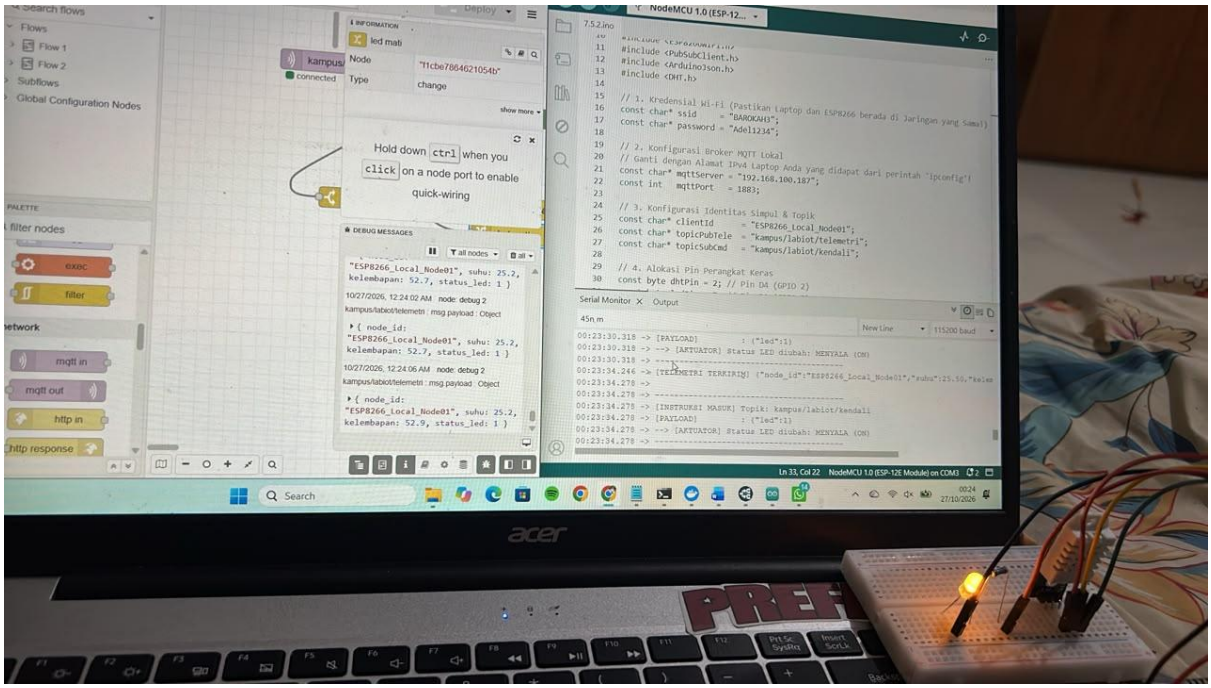


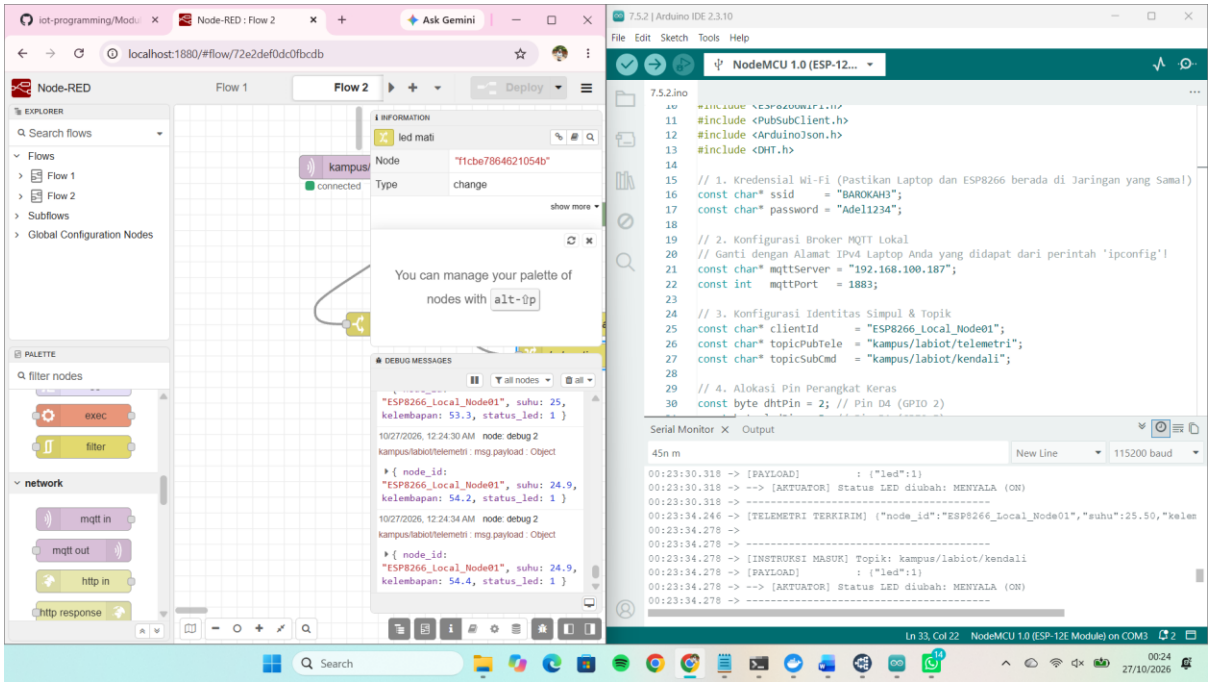
7.5.3 Praktikum 3: Integrasi Alur Data Telemetri Dua Arah pada Node-RED

Pada praktikum ini, kita akan menangkap dan membedah aliran data telemetri yang dikirimkan oleh NodeMCU menggunakan antarmuka kanvas Node-RED.



7.5.4 Praktikum 4: Implementasi Rule Engine Otomasi Aktuator Loop Tertutup





- **Latihan 3 (Implementasi Filter Histeresis pada Rule Engine):**
  Pada sistem kendali dua posisi (*Bang-Bang*), jika suhu berfluktuasi cepat di sekitar ambang batas $31.0°C$(misal: $31.05°C \rightarrow 30.95°C \rightarrow 31.02°C$), lampu LED indikator akan berkedip-kedip cepat (*flickering / hunting*) yang mengganggu stabilitas visual sistem.
  Rancang modifikasi alur Node-RED menggunakan **function node** berbahasa JavaScript yang menerapkan logika **Histeresis**:
  - LED baru menyala jika suhu $> 31.5°C$(msg.payload = {"led": 1}).
  - LED baru padam jika suhu mendingin hingga $<$ $30.0°C$(msg.payload = {"led": 0}).
    Tuliskan kode JavaScript yang Anda masukkan ke dalam *Function Node* tersebut!

JAWABANNYA:

```
// Mengambil status LED sebelumnya dari context (default 0/mati jika belum ada)
let ledState = context.get('ledState') || 0;


// Ambil nilai suhu dari msg.payload (asumsi masukan bertipe angka atau objek { temperature: value })
// Jika msg.payload berbentuk angka langsung:
let temp = typeof msg.payload === 'object' ? msg.payload.temperature : msg.payload;

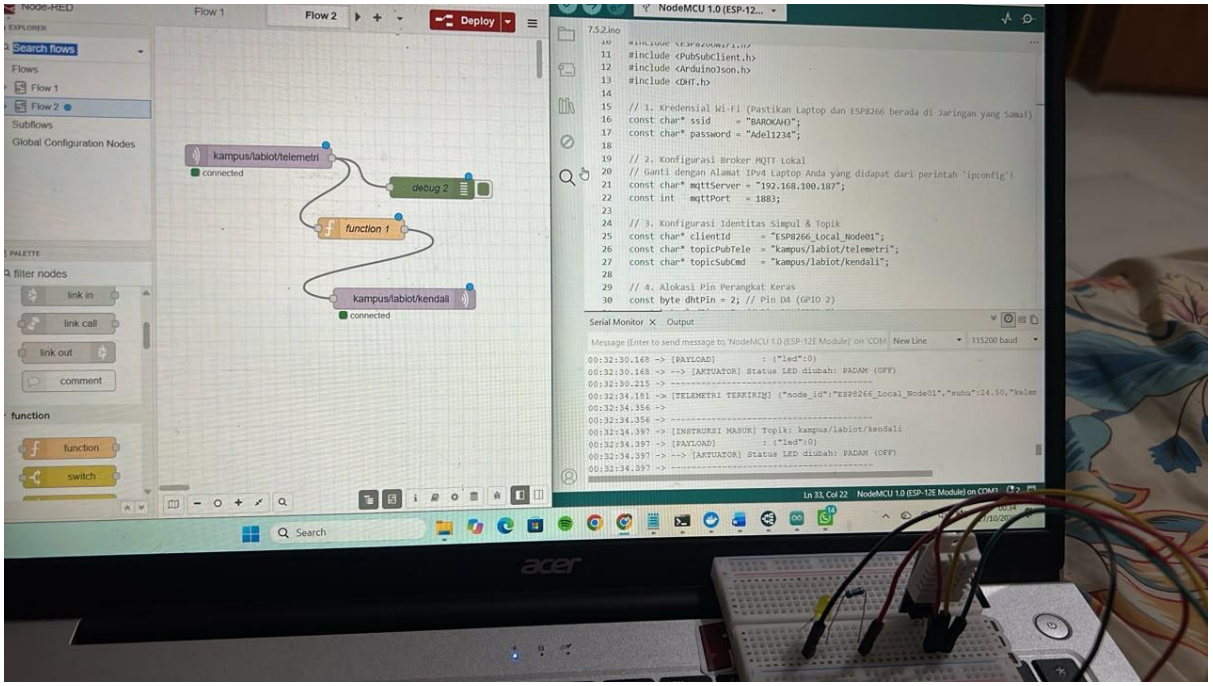

// Logika Histeresis
if (temp > 24.5) {
    ledState = 1; // Menyala jika suhu melebihi 24.5°C
} else if (temp < 23.0) {
    ledState = 0; // Padam jika suhu kurang dari 23.0°C
}
```

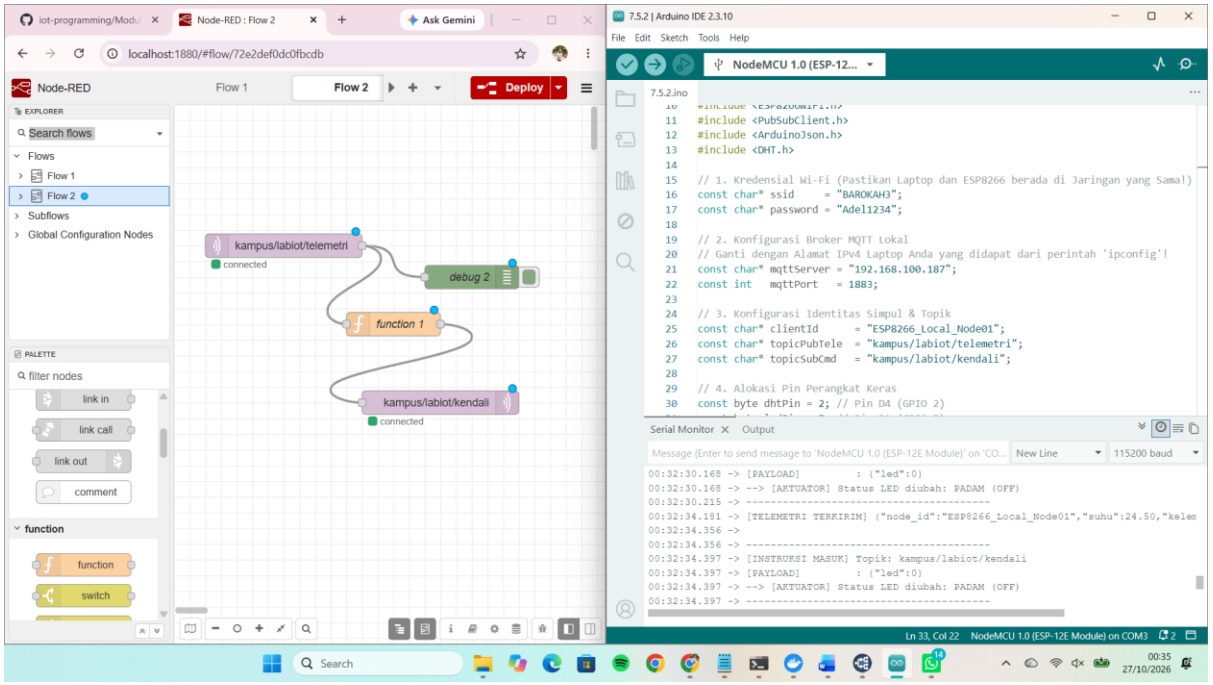
```
// Jika suhu berada di antara 23.0°C dan 24.5°C, nilai ledState tidak berubah (mempertahankan kondisi terakhir)

// Simpan kembali status LED terbaru ke context Node-RED
context.set('ledState', ledState);

// Format output sesuai kebutuhan
msg.payload = {
    "led": ledState
};

return msg;
```


Flow 1
Flow 2
Deploy
kampus/labiot/telemetri
connected
debug 2
function 1
kampus/labiot/kendali
connected
NodeMCU 1.0 (ESP-12...
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>
// 1. Kredensial Wi-Fi (Pastikan Laptop dan ESP8266 berada di Jaringan yang Sama!)
const char* ssid = "BAROKAH3";
const char* password = "Adel1234";
// 2. Konfigurasi Broker MQTT Lokal
// Ganti dengan Alamat IPv4 Laptop Anda yang didapat dari perintah 'ipconfig'!
const char* mqttServer = "192.168.100.187";
const int mqttPort = 1883;
// 3. Konfigurasi Identitas Simpul & Topik
const char* clientId = "ESP8266_Local_Node01";
const char* topicPubTele = "kampus/labiot/telemetri";
const char* topicSubCmd = "kampus/labiot/kendali";
// 4. Alokasi Pin Perangkat Keras
const byte dhtPin = 2; // Pin D4 (GPIO 2)
Serial Monitor
Output
00:32:30.168 -> [PAYLOAD] : {"led":0}
00:32:30.168 -> --> [AKTUATOR] Status LED diubah: PADAM (OFF)
00:32:34.181 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":24.50,"kelem
00:32:34.397 -> [INSTRUKSI MASUK] Topik: kampus/labiot/kendali
00:32:34.397 -> [PAYLOAD] : {"led":0}
00:32:34.397 -> --> [AKTUATOR] Status LED diubah: PADAM (OFF)


Node-RED : Flow 2
localhost:1880/#flow/72e2def0dc0fbcdb
Node-RED
Flow 1
Flow 2
Deploy
Search flows
Flows
Subflows
Global Configuration Nodes
filter nodes
link in
link call
link out
comment
function
switch
kampus/labiot/telemetri
connected
debug 2
function 1
kampus/labiot/kendali
connected
7.5.2 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
7.5.2.ino
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>
// 1. Kredensial Wi-Fi (Pastikan Laptop dan ESP8266 berada di Jaringan yang Sama!)
const char* ssid = "BAROKAH3";
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const char* mqttServer = "192.168.100.187";
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const char* topicPubTele = "kampus/labiot/telemetri";
const char* topicSubCmd = "kampus/labiot/kendali";
// 4. Alokasi Pin Perangkat Keras
const byte dhtPin = 2; // Pin D4 (GPIO 2)
Serial Monitor
Output
New Line
115200 baud
00:32:30.168 -> [PAYLOAD] : {"led":0}
00:32:30.168 -> --> [AKTUATOR] Status LED diubah: PADAM (OFF)
00:32:34.181 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":24.50,"kelem
00:32:34.397 -> [INSTRUKSI MASUK] Topik: kampus/labiot/kendali
00:32:34.397 -> [PAYLOAD] : {"led":0}
00:32:34.397 -> --> [AKTUATOR] Status LED diubah: PADAM (OFF)
Ln 33, Col 22
NodeMCU 1.0 (ESP-12E Module) on COM3
Search
00:35
27/10/2026